

EEG Preprocessing Pipeline — Emotiv EPOC X

Local-only MATLAB/EEGLAB pipeline that cleans Emotiv EPOC X recordings and plots their power spectra. Nothing leaves your computer at any point.

Quick start

Open MATLAB, then run these four lines:

```
cd('C:\Users\jacyp\OneDrive\EEG');  
  
addpath('src');  
  
cfg = setup_paths();    % finds EEGLAB, creates folders  
  
check_env();           % confirms toolboxes and plugins are present
```

Then process everything in [data/raw](#):

```
results = run_pipeline();
```

That is the whole thing. It imports every recording, cleans it, saves the cleaned version, computes the spectra, and writes a comparison figure.

Common variations

% Just one participant

```
results = run_pipeline('Subject', "126518");
```

% Just one condition

```
results = run_pipeline('Movement', "DOWN");
```

% Quick look at 3 files, saving a spectrum plot for each

```
results = run_pipeline('Limit', 3, 'PlotEach', true);
```

```
% Use a 50 Hz notch instead of the 60 Hz default
```

```
P = pipeline_config('doNotch', true, 'notchFreq', 50);
```

```
results = run_pipeline('Config', P);
```

Where things go

Folder	Contents
<code>data/raw/</code>	Your recordings (<code>.edf</code> , <code>.bdf</code> or <code>.csv</code>), exactly as exported. Never modified.
<code>data/processed/</code>	Cleaned data, one <code>.set</code> file per recording
<code>figures/</code>	Spectrum plots and the comparison figure
<code>results/</code>	<code>qc_summary.csv</code> plus the numeric spectra
<code>src/</code>	All the code
<code>toolboxes/</code>	EEGLAB

Start with `results/qc_summary.csv`. One row per recording, listing every decision the pipeline made: which channels were repaired, how many epochs were thrown away, whether ICA ran, and any warnings. The `usable` column is the one to check first — `0` means the recording was too noisy to trust.

What the code does

Read in this order if you want to follow it through:

File	Role
<code>setup_paths.m</code>	Finds EEGLAB, creates folders, returns all paths. Run first.

File	Role
<code>check_env.m</code>	Confirms required toolboxes and plugins are installed
<code>pipeline_config.m</code>	Every setting lives here. All thresholds, all on/off switches
<code>parse_recording_name.m</code>	Reads participant, condition and time out of a filename
<code>import_recording.m</code>	Entry point for loading. Routes EDF/BDF to <code>pop_biosig</code> , CSV to the Emotiv reader
<code>import_emotiv_csv.m</code>	Emotiv CSV → EEGLAB dataset, refusing to guess anything
<code>test_edf_roundtrip.m</code>	Proves the EDF path attaches the right data to the right channel name
<code>preprocess_recording.m</code>	The 8 cleaning stages
<code>compute_psd.m</code>	Turns cleaned data into a power spectrum
<code>plot_psd_stack.m</code>	Draws one recording
<code>compare_recordings.m</code>	Draws several side by side
<code>pick_three.m</code>	Chooses first / second-to-last / last
<code>run_pipeline.m</code>	Runs all of the above over every file
<code>make_test_fixture.m</code>	Builds fake data with a known answer, for testing

The 8 cleaning stages

1. **Reject bad segments** — cut out stretches where the signal has clearly broken down
2. **High-pass filter (1 Hz)** — remove slow drift. Raw Emotiv values sit near +4300 μV ; this is what removes that offset
3. **Low-pass filter (45 Hz)** — remove muscle activity and high-frequency noise
4. **Notch filter (optional)** — remove mains hum. Off by default; see below
5. **Re-reference** — express each channel against the average of all channels
6. **Artifact removal** — ASR for large sudden events, ICA + ICLabel for blinks and muscle
7. **Interpolate bad channels** — rebuild a broken electrode from its neighbours
8. **Epoch and reject** — cut into 2-second pieces, drop any still containing extremes

Every stage can be switched off in `pipeline_config.m`. Nothing is hard-coded into the pipeline body.

Reading the spectrum plots

Channels are stacked vertically, front of the head at the top. Left to right is frequency, 0–20 Hz. The height of each coloured band is how much power that channel carries at that frequency.

A bump around 8–12 Hz at the back of the head (O1, O2, P7, P8) is **alpha rhythm** — normal, and stronger with eyes closed.

The comparison figure puts three recordings side by side on **one shared scale**, so a column that looks lower really does have less power. If each were scaled to its own maximum, that difference would vanish.

Things worth knowing about this dataset

Findings from the 54 Harvard Dataverse recordings, which affect what the pipeline can honestly do:

- **The recordings are short.** Median 9 seconds, shortest 6, longest 74.
- **ICA is skipped on almost all of them.** ICA needs roughly 4,000 samples for 14 channels; a 9-second recording has about 1,150. The pipeline checks this and skips with an explanation rather than producing a decomposition that would look convincing and mean nothing. Only the 74-second recording qualified.
- **ASR is skipped for the same reason** — it needs a clean reference section within the recording, and there isn't one in 9 seconds.
- **The notch filter is off by default.** At 128 Hz the highest measurable frequency is 64 Hz, and the 45 Hz low-pass has already removed everything near 60 Hz. Turn it on if you record at 256 Hz.
- **2 of 54 recordings are flagged unusable** — too noisy to survive epoch rejection. They are reported, not silently dropped.

None of this is a fault in the code. It is what short recordings allow. Longer recordings — 60 seconds or more — would let ICA and ASR run, and those are the two stages that do the heavy lifting on blink and muscle artifacts.

Input formats

Drop `.edf`, `.bdf` or `.csv` files into `data/raw/` — `run_pipeline()` picks up all three and routes each to the right reader. EDF/BDF go through `pop_biosig`; CSV goes through the EmotivPRO reader.

Channels are matched by name, never by position. An EDF that stores its channels in a different order, or labels them `EEG AF3` instead of `AF3`, is reordered into the correct EPOC X montage on import. This is not cosmetic: taking channels 1–14 positionally would produce a scrambled montage that runs without error and makes every figure wrong. To verify it on your machine:

```
test_edf_roundtrip(cfg);
```

That writes an EDF with deliberately shuffled channels, reads it back, and checks the right signal ends up under the right name.

Testing without real data

`make_test_fixture.m` writes a synthetic file with a known answer: a 10 Hz peak at the back of the head, blinks at the front, mains hum, drift, and one dead channel.

```
fixture = make_test_fixture(cfg);
```

```
EEG = import_emotiv_csv(fixture, cfg);
```

```
S = compute_psd(EEG, pipeline_config());
```

```
plot_psd_stack(S, cfg);
```

If the peak does not land at 10 Hz, something is wrong with the analysis rather than with the data. Delete the file from `data/raw/` afterwards — it is not a recording.

If something goes wrong

"EEGLAB not found" — EEGLAB isn't in `toolboxes/`. See `SETUP.md`.

"Channel location file not found" — `data/emotivX_channels_location.ced` is missing. Download link is in `SETUP.md`.

"Could not determine the sampling rate" — the file has no readable header. Pass it explicitly, but only if you know it for certain: `import_emotiv_csv(file, cfg, 'SampleRate', 128)`.

A recording fails — the batch continues and reports which one and why at the end. One bad file never stops the run.

Requirements

- MATLAB R2021a or later, with Signal Processing and Statistics toolboxes
- EEGLAB, with the BIOSIG, firfilt, clean_rawdata and ICLabel plugins

`check_env()` confirms all of this and names anything missing.